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Ultrashort pulse laser welding – a new approach for high-stability bonding of different glasses

Sören Richter[a,*], Stefan Nolte[a,b], Andreas Tünnermann[a,b]

[a]*Institute of Applied Physics, Abbe Center of Photonics, Friedrich-Schiller-Universität Jena, Max-Wien-Platz 1, 07743 Jena, Germany*

[b]*Fraunhofer Institute for Applied Optics and Precision Engineering, Albert-Einstein-Straße 7, 07745 Jena, Germany*

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Abstract

Ultrashort laser pulses at high repetition rates may overcome the problems of conventional bonding methods. While focusing in transparent materials, the laser energy is deposited locally in the focal volume due to nonlinear absorption processes. Heat accumulation of successive laser pulses at high repetition rates leads to localized melting of the material. If the focus is set at the interface of two transparent samples, the subsequent resolidification finally yields strong and robust bonds. Using optimized processing parameters, we achieved a breaking strength, which is comparable to the bulk material.

Keywords: welding; heat accumulation; transparent material; breaking strength; ultrashort pulses

1. Introduction

The combination of excellent chemical, mechanical and optical properties with well-known processing methods makes glass the most important material for the realization of optical systems. The development of microoptical devices involves the integration of several materials into complex structures. However, the stable bonding of different glasses still remains a technological challenge.

The different conventional bonding techniques are subject to distinct disadvantages. The simplest bonding technique for surfaces is optical contacting. For this adhesive–free process samples with a very

* Corresponding author. Tel.: +49-3641-947820 .
E-mail address: stefan.nolte@uni-jena.de .

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high surface quality are required. The molecular attraction between the surface atoms of the samples lead to a bond strength of a few kPa. To increase the bond strength, several bonding techniques have been developed [1]. One method comprises the usage of adhesives or glue [2]. However, glue is prone to aging and gas emission. Another option is low pressure plasma surface activation, called Direct Bonding [3,4], which yields a bonding strength of several MPa but requires numerous processing steps.

Most common bonding techniques include an annealing step to increase the bonding strength [5]. Unfortunately, the heating of both samples result in internal stress while bonding glasses with different thermal expansion coefficients. Furthermore, with the exception of adhesive methods, none of the conventional techniques allows for the selective bonding of specific areas of the interface while leaving the remaining area unaffected.

For the bonding of transparent materials with an opaque material laser radiation may be used [6,7]. The linear absorption at the opaque material leads to melting of the interface (see figures 1 a-c). This technique enables e.g. the bonding of glass on metals [6] (figure 1b). While using an absorption layer, even the bonding of two transparent materials can be achieved (figure 1c). However, this technique is not suited for the bonding of two transparent materials without any intermediate layer (figure 1d).

This limitation can be overcome by using ultrashort laser pulses as a local heat source. Due to nonlinear absorption processes, the laser pulse energy is deposited solely in the focal region, which leads to local melting of the sample (figure 1e) [8]. If the laser focus is located at the interface of two samples, the resolidification of the molten material forms strong bonds [9,10] between the samples. The accessible breaking strength may be enhanced by using high repetition rate laser system [11]. If the time between two laser pulses is shorter than the time for heat diffusion out of the focal volume (about 1 µs in fused silica), heat accumulation of successive pulses occurs. Thereby even the material surrounding the focal spot is molten [12,13].

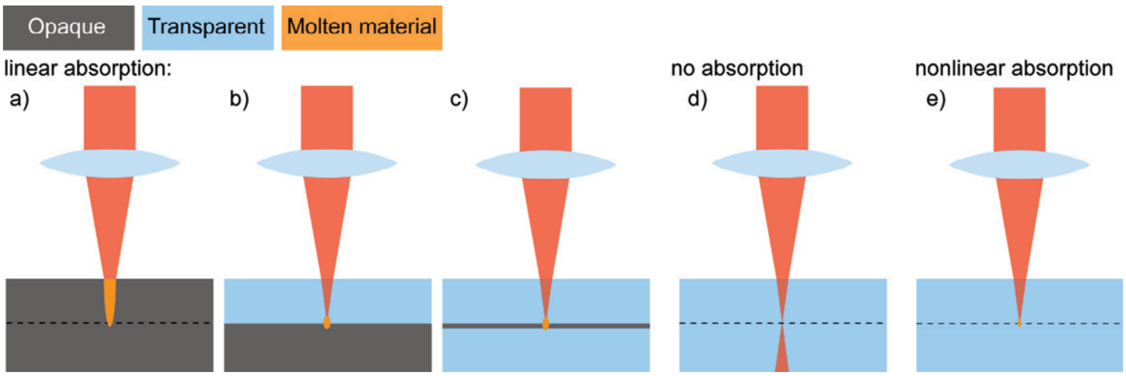

Fig 1. Comparison of the laser induced melting of different material combinations using linear absorption (a – c) and nonlinear absorption (e). While linear absorption requires an opaque material, nonlinear absorption occurs even in transparent materials at high energy densities. The interface of two samples is marked as a dashed line

The application of ultrashort laser pulses at high repetition rates as local heat source offers several advantages. As no annealing step is required to strengthen the bonds, materials with widely different thermal expansion coefficients can easily be joined [14,15]. Since the heat affected zone has only a characteristic dimension of a few µm, the induced thermal stress in the material is minimized. In addition, tailored bonding for custom designed applications becomes easily possible, as the laser focus may readily be moved along the interface.

2. Experimental setup

For the bonding experiments, we used a femtosecond oscillator delivering pulses at a wavelength of 1030 nm, at pulse energy of 500 nJ, a pulse duration of 450 fs and a repetition rate of 9.4 MHz (*Amplitude, t – Pulse 500*). We conducted our welding experiments at a wavelength of 515 nm, which was generated using an LBO crystal. An acoustic optic modulator and polarization optics were used to change the repetition rate and pulse energy, respectively. To focus at the interface of two 6.4 mm thick samples, we used an aspheric lens with a focal length of 4.5 mm (NA 0.55). More details about the experimental setup can be found in [11].

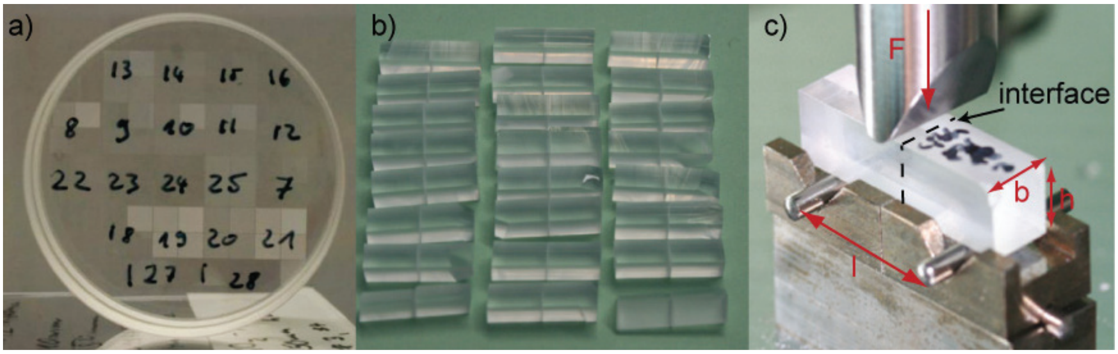

Fig. 1. Measurement of the breaking strength of laser welded samples. The areas with differently inscribed modifications (a) where cut into rectangular rods (b). The breaking strength σ was finally measured with a three point bending test (c)

Before the welding experiments, we optically contacted our samples. After the inscription of the different welding seams (figure 2a), we cut the bonded samples into rectangular rods (figure 2b). To determine the breaking strength of the weld, we conducted a three point bending test (figure 2c). The breaking strength σ can then be calculated from the force F required to break the bonded samples apart with the formula

$$\sigma = \frac{3Fl}{2bh^2} \qquad (1)$$

Here, b and h refer to the edge lengths of the sample cross-section and l is the distance between the two bearings (see figure 2 c). The cutting process induces defects and scratches along the outer surface of the rod. These defects exhibit the main error source of the bending test, which is for bulk material in the range of about 25%. For bonded samples, the original interface acts as a predetermined breaking zone. This reduces the influence of scratches and defects, yielding an error of about 15%. We conducted our experimental investigations with different glasses: fused silica (Schott Q1), borosilicate glass (Schott Borofloat 33) and ULE^{TM} (Corning).

3. Results

3.1. Laser –induced material modifications in transparent materials

Figure 3 a) shows exemplary side views of the laser induced material modifications in two bonded borosilicate samples. The heat affected zone has a characteristic elliptical shape [12, 13, 15], whose size and shape is defined by the applied laser power [16]. While bonding different glasses (e.g. borosilicate and fused silica; figure 3 b)) the modification appears elliptically, too. However, the width of the welding is strongly dependend on the thermal conductivity and the softening point of the material. The thermal conductivity of borosilicate is 1.2 W/m/K , whereas fused silica depicts 1.38 W/m/K. In addition, the softening point of fused silica (1585°C) is much higher, than borosilicate (985°C). Thus, the width of the wealding seams in fused silica is mich smaller than in borosilicate.

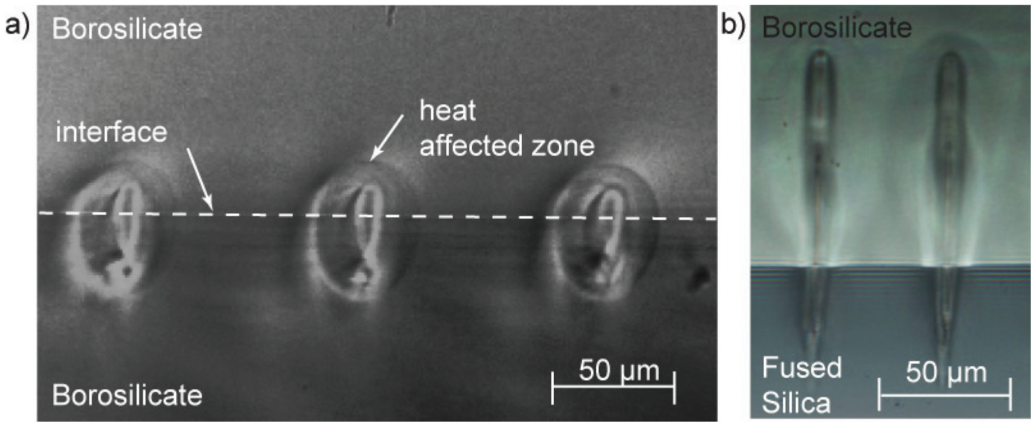

Fig. 1. Heat affect zone induced by ultrashort laser pulses at high repetition rates at the interface of different glass samples. The laser irradiates the samples from the top

3.2. Welding and breaking strength of different glasses

In order to achieve a high breaking strength, we conducted our experiments in the heat accumulation regime. Our first measurements of the bulk material of fused silica result in a breaking strength of about 70 MPa, which is in good agreement with results shown by *Heraeus* [17]. If bulk material is structured with the same parameters as for the laser welding process the breaking strength decreases to 58 MPa.

In a next step, we measured the breaking strength of laser-welded fused silica samples with respect to the actual amount of molten material at the interface. Our investigations show that the breaking strength generally increases with the molten area at the interface (Figure 4 a) [11]. If the welding is done as continuous lines at the interface, the breaking strength at first increases linearly with the molten area and then saturates at about 25 N/mm² (40% of the bulk material). In contrast, if isolated welding spots are used, a breaking strength of 54 N/mm² is achieved. This is about 75% of the breaking strength of the bulk material and corresponds to the results obtained with laser structured bulk material. The different breaking strength obtained with different welding geometries may be explained with the ratio between boundaries of the welding seams and the enclosed molten material. Each border forms a resistance to the crack propagation. Thus, a crack may easily propagate along a continuous line as soon as the weld is damaged. However, at isolated spots the number of borders is significantly increased and hence a higher breaking strength is required to crack these welds. Generally, the breaking strength saturates when about 60% of the interface is molten. Under these conditions, a crack rather propagates in a plane parallel to the interface, than at the original interface (Figure 4 b). Although, the welds establish new bonds between the glass samples (left side of Figure 4 b), the welding process induces some tensions in the material, too. If the distance between the molten spots is mall enough, the induced tensions overlap and form a new predetermined breaking plane. Then, a crack propagates rather at this breaking plane in the bulk material than in the original interface.

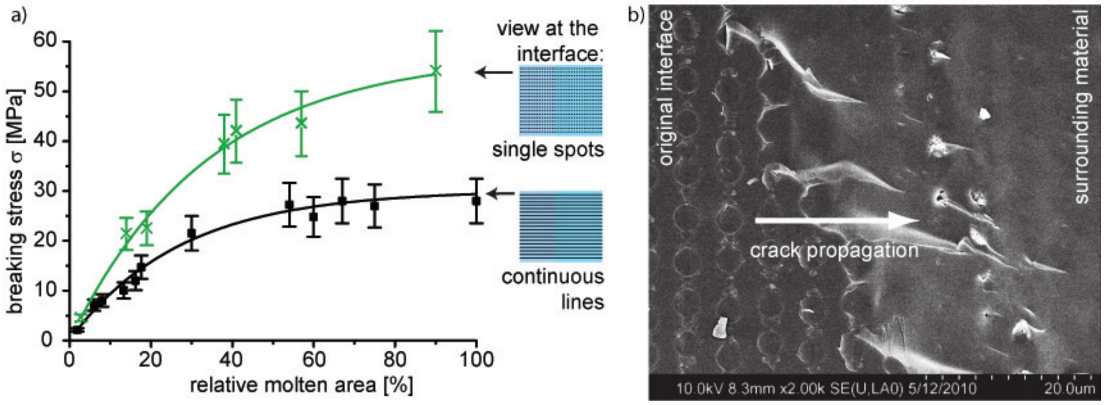

Fig. 4. (a) The breaking stress of laser welded fused silica increases with the molten area. Isolated weldings spots exhibit higher breaking strengths than continuous lines; (b) If the welding seams are close together, the induced tensions overlap and a crack propagates rather along these tensions fields, than in the original interface

We did the same measurement with different glasses. At first we conducted experiments with a borosilicate glass. In comparison to fused silica, borosilicate glass has a much higher amount of network modifiers. This leads to a reduction of tensions and disruptions induced by the laser welding process [18]. Furthermore, borosilicate has a relatively low softening point (985°C), therefore the size of the molten volume is increased. With optimized processing parameters we achieved a maximal breaking strength of 95% of the bulk material (113 MPa). However, even for this very high breaking strength we observed the same behavior for the crack propagation as described above for fused silica.

In addition, we worked with ULE glass. Here, intensive laser irradiation leads to darkening of the glass, due to the reduction of TiO_2 content to Ti_2O_3, TiO and Ti [19]. The reduction of TiO_2 occurs at temperatures above 1500°C, which is a similar value as the softening point of ULE (1490°C). The darkening hinders the formation of strong bonds. Therefore, the optimized welding parameters have to be adapted to exhibit a large molten volume with a relative small amount of transformed TiO_2. While using relative low pulse energies below 100 nJ and a repetition rate of 4.7 MHz, we achieved a maximum breaking strength of 50 MPa, which is about 60% of the bulk material [20].

Finally, to underline the advantages and versatility of this process we welded different glass samples. Due to the different thermal expansion coefficients, annealing of the samples would lead to intensive tensions. Therefore common bonding techniques are not able to bond different materials. However, several groups [14,15,20,21] have already demonstrated the possibility of welding even different materials using ultrashort laser pulses. One combination of different glasses was shown in Figure 3 c), where we welded fused silica with borosilicate glass. To determine the accessible breaking strength we used again the three point bending test. Figure 5 summarizes our first results. The light grey bars show the breaking strength of the bulk material, which we obtained as an average value of about 20 bulk bending measurements. As we changed the sawing blade of the glass saw, we obtained a slightly different bulk value for fused silica, as stated above. The grey bars in front of the light grey ones show the obtained breaking strength for homogenous material combinations. Furthermore, we welded different glasses together and achieved breaking strengths which are comparable to the homogenous combinations. We used black bars to indicate our best results. For example, the combination of SiO_2 and borofloat yields a maximal breaking strength of 75 MPa. For the welding of fused silica on ULE, we had to regard the darkening of the ULE sample and the high softening point of fused silica, too. Here, we achieved a breaking strength of almost 60% of bulk ULE or fused silica, respectively.

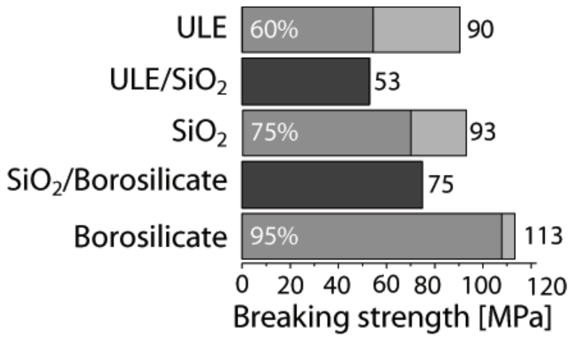

Fig. 5. Summary of the obtained breaking strengths for different glasses as well as different glass combinations. The light grey bars show the breaking strength of the bulk material, whereas the grey bars indicate to the maximal breaking strength of homogeneous glass combinations. The dark grey bars state the welding strength of different glass combinations

4. Conclusion

Ultrashort laser pulses at high repetition rates offer the possibility to realize strong bonds between different glasses. If the laser focus is placed at their interface the combined action of nonlinear absorption processes and heat accumulation leads to welding of the samples. We have investigated the dependence of the accessible breaking strength on the laser parameters for different glasses. To this end, we used a three point bending test. By optimizing the processing parameters we were able to obtain a breaking strength almost as high as the bulk material. These high values of the breaking stress demonstrate the potential of this technique for the flexible realization of integrated microoptical devices.

Acknowledgements

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